

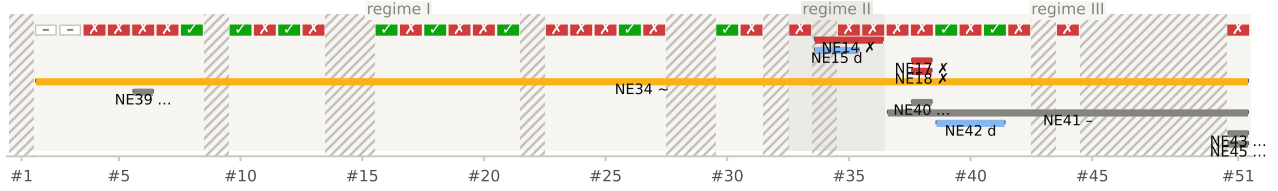
H36: A reorganization alarm: susceptibility and multi-information peak at transitions

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. Multi-information and susceptibility of an LLM swarm barely mark its reorganizations: a third of goal changes are caught, all through content, and a plain topic-shift detector does far better.

level: hypothesis · pipeline checks 84% · updated 2026-10-04



Goal periods tested (10 supported, 26 failed, 1 mixed, 2 descriptive, 4 pending, 3 n/a). ✓ supported ✗ failed ~ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

Each non-holdout village day (279) we computed three fluctuation statistics among agents: multi-information, susceptibility (variance of the collective mode) and a heat-capacity analogue (variance of the alignment energy). They were computed for 1-min activity spins, 4-state behavior and 30-min content vectors, as excesses over shuffled surrogates. The pre-registered alarm fires when their trailing z-score reaches 2. Synthetic check: content-coupling switches are caught (AUC 0.93), activity-coupling ones only partly (0.77), day-boundary field steps not at all. **Real data, mixed:** the alarm caught 10 of 33 goal changes within a day, at a 10% placebo-window false-alarm rate (AUC 0.68, random-date $p = 0.03$). The signal is entirely content: activity is at chance once roster and day-length changes are excluded, and ~40% of the content signal is a within-day drift. The plain day-to-day shift of the mean content vector scores AUC 0.95. Scaffold changes: 1/13. Holdout not run.

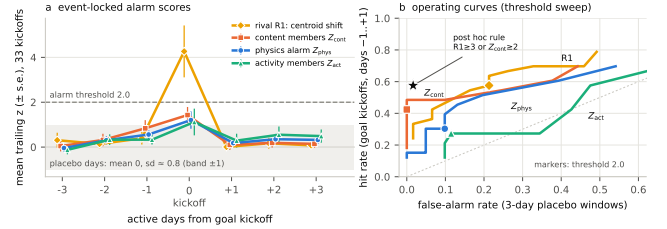
PHYSICAL MODEL

Models 01 (inverse Ising) and 11 (vector spins). $s_i(t) = \pm 1$: agent i active in minute t ; $M = \sum_i s_i$; $\hat{s}_i(w)$: whitened, unit content vector of agent i in window w . VR: susceptibility as a variance ratio (1 for independent agents); c : heat-capacity analogue with the uniform alignment energy E ; I_G : Gaussian multi-information ($R =$ correlation matrix); $e(w)$: content alignment energy. Every statistic X enters as observed minus surrogate mean, then a trailing z over the previous 10 days.

$$\text{VR} = \frac{\text{Var}_t M}{\sum_i \text{Var}_t s_i}, \quad c = \frac{\text{Var}_t E}{n}, \quad E = -\frac{1}{n} \sum_{i < j} s_i s_j$$

$$I_G = -\frac{1}{2} \log \det R, \quad e(w) = -\langle \hat{s}_i(w) \cdot \hat{s}_j(w) \rangle_{i < j}$$

$$z_X(d) = \frac{X(d) - \text{med}_{10} X}{\text{sd}_{10}^{\text{trim}} X / c_{10}}, \quad Z_{\text{phys}} = \langle Z_I, Z_X, Z_C \rangle \geq 2$$



Observables. (a) Mean alarm scores around 33 goal kickoffs. The physics alarm (blue) and its content members (orange) rise on the kickoff day but stay below threshold on average. The activity members (aqua) barely move. The content-centroid shift (yellow) jumps. (b) Hit rate against placebo-window false-alarm rate as the threshold is swept; markers sit at threshold 2. The star is the post hoc rule ($R1 \geq 3$ or $Z_{\text{cont}} \geq 2$): 58% hits at 2% false alarms, in-sample.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

As posed, a weak alarm. Swarm-wide fluctuation statistics do not light up when an LLM swarm reorganizes. Activity fluctuations react to who is present and for how long; content fluctuations react mostly to the within-day settling after a new goal. A first-moment detector, “did the topic move since yesterday?”, is cheaper and much better. The physics adds one usable thing, pending the holdout: the content-fluctuation score is a good false-alarm filter beside the topic-shift detector. It matches the shift detector’s hit rate at about a tenth of the false alarms. Scaffold changes, the events an operator most wants flagged, stay invisible to all of these daily statistics. An honest negative for “susceptibility peaks at transitions”.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 goal hit $\leq 40\%$, AUC < 0.65	hit 0.30 (10/33), window FAR 0.10, AUC 0.68 [0.56, 0.79], random-date p 0.03	mostly held; card rule: mixed
P2 content $>$ activity	AUC 0.77 vs 0.55 (0.47 without roster/hours changes)	held
P3 rival R1 wins	R1 AUC 0.95; Δ AUC $[-0.39, -0.13]$	held
P3 adding Z_{phys} to R1 gains < 0.1	$+0.15$ hits, $+0.10$ window FAR	failed
P4 merge raises activity χ, I	05-04: $z = -0.6$; splits lower ($-3.0, -0.8$)	mostly failed
P5 NE14 alarms; other scaffold at chance	03-24 silent; 1/13 vs FAR 0.10	half
P6 stalls cause false alarms	0 unmasked false alarms on placebo days	failed
P7 per-day FAR 0.03–0.12	FAR 0.033; Mondays 0.09 vs 0.02	held
P8 no day -1 signal	3/10 hits on day -1 ; AUC 0.69 vs Fridays (post hoc)	partly

GOAL PERIODS IN DETAIL

G04 $\times$ failed
G05 $\times$ failed
G06 $\times$ failed
G07 $\times$ failed
G08 $\checkmark$ supported
G10 $\checkmark$ supported
G11 $\times$ failed
G12 $\checkmark$ supported
G13 $\times$ failed
G16 $\checkmark$ supported
G17 $\times$ failed
G18 $\checkmark$ supported
G19 $\times$ failed
G20 $\times$ failed

+32 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Daily resolution only, so mid-day room events cannot be timed. There are just 61 placebo days, none in regime II; zero false alarms bounds the per-day FAR at about 5%. Kickoffs are mostly Mondays, and only 11 placebos are. The activity channel reacts to roster and day-length changes. About 40% of the content signal is a within-day drift after the kickoff. One embedding model. The operator rule was chosen post hoc among 7 candidates (in-sample 58% hits at 2% window false alarms); it is frozen as C3 for the holdout. R1 is missing after held-out days. H38's outage table was used while H38 was still running.

ROUND 2

Run the holdout confirmation (C1–C5) after commit and the reuse disclosure. Then build an intraday (30–60 min) version and statistics that don't move when the roster or day length changes. For scaffold changes, which no daily swarm statistic sees, look at action-mix change points.

- **Essence.** a cheap, log-computable alarm for structural change in a swarm (goal, roster, room, scaffold) with known hit and false-alarm rates; round 1 says content carries it, activity fluctuations don't, and a first-moment topic-shift detector beats the physics statistics.
- **H36-R1** Holdout confirmation.
- **H36-R2** Intraday alarm.
- **H36-R3** Sampling-invariant activity statistics.
- **H36-R4** Embedding swap.
- **H36-R5** Scaffold detector.

Second figure. Synthetic validation at village sampling (30 runs per scenario; AUC of the switch window against placebo windows). Coupling switches in content are recovered; in activity only partly; day-boundary field steps and goal switches are invisible to the fluctuation statistics but obvious to the first-moment rival R1.

